⚡ Keysight E36300 PSU Voltage Sequencing

RAG Conversation: E36300 Series Power Supply Unit (PSU) Programming

❓ **User Request**

Refer to E36300 Series PSU manual, create minimal code for:

- Set Vout to 3V
- After 1 second → Set to 5V
- Max current: 0.5A

✅ RAG Response

Based on the Keysight E36300 Series Programming Guide (9018-04577) and User's Guide, here is the minimal Python code to achieve your request.

Key Technical Details from Manual

- **Command Structure:** The `APPLY` command handles both voltage/current setting and sequencing in one call for this series.

  > **Syntax:** `APPLy <Channel>,<Voltage>[,<Current>]`

- **Sequencing (Step 1 → Step 2):** For simple step changes in E36300, sending two `APPLY` commands with a delay between them is the standard approach unless using the advanced `LIST` subsystem (which requires pre-loading steps).
- **Channel:** You must specify the channel (e.g., `CH1`, `P6V`, `P25V`). Assuming CH1 for this example.
- **Termination:** Explicitly set to `\n` as per SCPI standards and requirements.

Minimal Python Code (PyVISA)

```python
import pyvisa
import time

# 1. Setup VISA Resource Manager
rm = pyvisa.ResourceManager()

# 2. Define Instrument Address (Replace with actual IP/Port)
# Example: TCPIP0::192.168.1.50::INSTR
resource_name = "TCPIP0:::::SOCKET"

try:
    # Open connection
    inst = rm.open_resource(resource_name, write_termination='\n', read_termination='\n')

    print("--- E36300 Sequence Setup ---")

    # 3. Handshake & Reset (Best Practice)
    print("Identifying instrument...")
    idn = inst.query("*IDN?").strip()
    print(f"Found: {idn}")

    print("Resetting to defaults...")
    inst.write("*RST")

    print("Clearing status registers...")
    inst.write("*CLS")

    # 4. Execute Sequence
    channel = "CH1"

    # Step A: Set Voltage to 3V, Current Limit to 0.5A
    cmd_a = f"SOURce:{channel} VOLTage 3\nSOURce:{channel} CURRent 0.5\nOUTPut ON"
    print(f"Sending Step 1 (3V @ 0.5A): {cmd_a}")
    inst.write(cmd_a)

    # Wait 1 second before changing voltage
    time.sleep(1.0)

    # Step B: Set Voltage to 5V, Current Limit remains 0.5A (or re-specify if needed)
    cmd_b = f"SOURce:{channel} VOLTage 5\nSOURce:{channel} CURRent 0.5\nOUTPut ON"
    print(f"Sending Step 2 (5V @ 0.5A): {cmd_b}")
    inst.write(cmd_b)

    # Optional: Verify final state
    v_meas = float(inst.query("MEASure:VOLTage?").strip())
    c_meas = float(inst.query("MEASure:CURRent?").strip())
    print(f"Verification - Measured Vout: {v_meas}V, Iout: {c_meas}A")

except Exception as e:
    print(f"Error occurred: {e}")
finally:
    # 5. Cleanup
    if 'inst' in locals():
        inst.close()
```

Explanation of Commands Used

Command	Description
`*RST`	Resets the instrument to factory defaults (clears previous settings)
`SOURce:CH1 VOLTage <val>`	Sets the output voltage for Channel 1
`SOURce:CH1 CURRent <val>`	Sets the maximum current limit (OCP) to 0.5A
`OUTPut ON`	Enables the channel after setting parameters (required after setting voltage/current if previously off)
`time.sleep(1.0)`	Implements the 1-second delay between steps
`MEASure:VOLTage?` / `MEASure:CURRent?`	Queries the actual measured values to verify load is powered correctly (Safety Check)

Command Flow Breakdown

Step 1: Initialize

- ***IDN?** — Identify the instrument (verify connection)
- ***RST** — Reset to factory defaults
- ***CLS** — Clear status registers

Step 2: Configure & Enable (First Voltage)

- **SOURce:CH1 VOLTage 3** — Set to 3V
- **SOURce:CH1 CURRent 0.5** — Set current limit to 0.5A
- **OUTPut ON** — Enable output

Step 3: Wait 1 Second

- **time.sleep(1.0)** — Allow settling time

Step 4: Change Voltage (Second Voltage)

- **SOURce:CH1 VOLTage 5** — Set to 5V
- **SOURce:CH1 CURRent 0.5** — Maintain current limit
- **OUTPut ON** — Ensure output stays enabled

Step 5: Verify & Cleanup

- **MEASure:VOLTage?** — Query actual voltage reading
- **MEASure:CURRent?** — Query actual current reading
- **Close connection** — Cleanup resources

Safety Notes

- ⚠️ Voltage Limits: Ensure your load can handle a step from 3V to 5V. Sudden voltage changes can stress components.
- ⚠️ Current Limit: The limit is set to 0.5A for both steps. If the load draws less than this, it will run in **Constant Voltage (CV) mode**. If it tries to draw more, it will trip into **Constant Current (CC) mode** at 0.5A.
- ⚠️ Grounding: Ensure the PSU chassis is grounded before connecting loads, especially when switching voltages rapidly.

Expected Console Output

```
--- E36300 Sequence Setup ---
Identifying instrument...
Found: KEYSIGHT,E36300,SN1234567890,FW1.0

Resetting to defaults...
Clearing status registers...
Sending Step 1 (3V @ 0.5A): SOURce:CH1 VOLTage 3
SOURce:CH1 CURRent 0.5
OUTPut ON

[Wait 1 second...]

Sending Step 2 (5V @ 0.5A): SOURce:CH1 VOLTage 5
SOURce:CH1 CURRent 0.5
OUTPut ON

Verification - Measured Vout: 5.000V, Iout: 0.000A
```

Common Variations

If you need to control a different channel:

- `P6V` — 0–6V, 5.15A channel
- `P25V` — 0–25V, 1.03A channel
- `N25V` — −25V to 0V, 1.03A channel

Simply replace `CH1` with the desired channel in the code above.

Advanced: Using APPLY Command (Condensed)

The E36300 also supports the condensed `APPLY` syntax:

```python
# Alternative: Using APPLY command
inst.write("APPLY CH1,3,0.5")       # Step 1: 3V, 0.5A
time.sleep(1.0)
inst.write("APPLY CH1,5,0.5")       # Step 2: 5V, 0.5A
```

This is shorter but less explicit. Both methods work identically on the E36300.